

Date : 14th Sep 2020, Monday

Time : 03:00pm To 05:00pm

Session : Evening

Sub: Computer based numerical & statistical

Course No : PS04FIIT01

Total marks : 70

Q – 1 Multiple Choice Question

[10]

- i) From the following _____ method is the best method to obtain root of equation $f(x)=0$.
a) False position b) Bisection
c) Newton's Raphson d) None of Them
- ii) True error is defined as
a) Present Approximation – Previous Approximation
b) True Value – Approximate Value
c) $\text{abs}(\text{True Value} - \text{Approximate Value})$
d) $\text{abs}(\text{Present Approximation} - \text{Previous Approximation})$
- iii) The number 0.01850×10^3 has _____ significant digits Referential
a) 3 b) 4 c) 5 d) 6
- iv) To estimate the value of dependent variable x for a given value of independent variable y , the process is known as _____.
a) Extrapolation b) Inverse Interpolation
c) Interpolation d) Polynomial
- v) _____ is called the backward difference operator.
a) Δ b) ∇ c) $\emptyset$ d) $\cup$
- vi) y depends on x can be written as _____.
a) $f(x)$ or y_x b) $f(xy)$ c) $f(yx)$ d) None
- vii) We can find solution of system of linear, algebraic equations using _____.
a) Newton-Raphson method b) Bisection method
c) Gauss-Seidel method d) None
- viii) The system of linear equation $AX = B$ can be solved by matrix inversion method only if _____.
a) $A \neq 0$ b) $|A| \neq 0$ c) $|A| = 0$ d) A is symmetric
- ix) Forecasts _____.
a) become more accurate with longer time horizons b) are rarely perfect
c) are more accurate for individual items than for groups of items d) all of the above
- x) Gradual, long-term movement in time-series data is called _____.
a) seasonal variation b) cycles
c) trends d) exponential variation

Q – 2 Fill in the Blanks and True – False

[08]

- i) The lagranges method is not iterative method. (True / False)
- ii) For real root of an equation $x^3-2x-5=0$ the root lies between _____.
- iii) Theile defines Extrapolation as "The art of reading between the lines of a table."
(True / False)
- iv) All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _____.
- v) Rate of change of distance with respect to time represents Acceleration.
(True / False)
- vi) The system of linear equation $AX=B$ is said to be non-homogeneous if _____
- vii) A _____ of a time series attached to long-term variations is term as Secular trend
- viii) A time series consist of four components. (True / False)

Q – 3 Short Answer attempt any ten (Each carry 2 marks)

[20]

- i) Define Relative error and absolute error.
- ii) Use the False Position method to obtain initial solution of $x^3 - 9x + 1 = 0$
- iii) Use the secant method to obtain initial solution of the equation $x^3 - 5x + 3 = 0$.
- iv) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- v) Define Interpolation.
- vi) Write algorithm for Backward difference table
- vii) List the component of Time series.
- viii) Solve the system of equation: $2x + 3y = -10$
 $-x + 4y = -4$
- ix) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- x) What is Time Series?
- xi) What do you mean by Secular trend?
- xii) What do you mean by random or irregular Variation?

Q – 4 Attempt any four (Each carry 8 marks)

[32]

- i) Write algorithm for Regular Falsi method.
- ii) Find the root of equation $2^x - x - 3 = 0$ correct up to four decimal places using Iterative method. (Take $a = -3$).
- iii) The following table gives the census population of a town for the years 1931 to 1971. Estimate the population for the year 1965 by using an appropriate interpolation formula.

Year	1931	1941	1951	1961	1971
Population	46	66	81	93	101

- iv) Given a function in the form of a table as

x	2.0	3.0	4.0
y(x)	6.6	9.2	8.6

Interpolate the value of $y(x)$ using Lagrangian polynomial at $x = 2.8$ and $x = 3$.

- v) Explain the Gauss – Sedel method for solution of system of linear equation.
- vi) The distance (s) covered by a car in given time (t) is given in the following table:

Time(minutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

Determine the speed of the car at $t = 13$ minutes.

- vii) Obtain seasonal indices using simple average method.

Year	Q-I	Q-II	Q-III	Q-IV
1990	30	81	62	119
1991	33	104	86	171
1992	42	153	99	221
1993	56	172	129	235

- viii) Obtain seasonal indices using ratio to moving average.

Year	Q-I	Q-II	Q-III	Q-IV
1990	30	40	36	34
1991	34	52	50	44
1992	40	58	54	48
1993	54	76	68	62